

Collaborative Avatar Platform for Collective Human Expertise

Takayoshi Hagiwara
Keio University Graduate School of
Media Design
Japan
hagiwara@kmd.keio.ac.jp

Takumi Katagiri
Nagoya Institute of Technology
Japan
t.katagiri.753@nitech.jp

Hikari Yukawa
Nagoya Institute of Technology
Japan
yukawa.hikari@nitech.ac.jp

Itsuki Ogura
Nagoya Institute of Technology
Japan
i.ogura.373@nitech.jp

Ryohei Tanada
Nagoya Institute of Technology
Japan
r.tanada.550@nitech.jp

Takumi Nishimura
Nagoya Institute of Technology
Japan
clp13218@nitech.jp

Yoshihiro Tanaka
Nagoya Institute of Technology
Japan
tanaka.yoshihiro@nitech.ac.jp

Kouta Minamizawa
Keio University Graduate School of
Media Design
Japan
kouta@kmd.keio.ac.jp

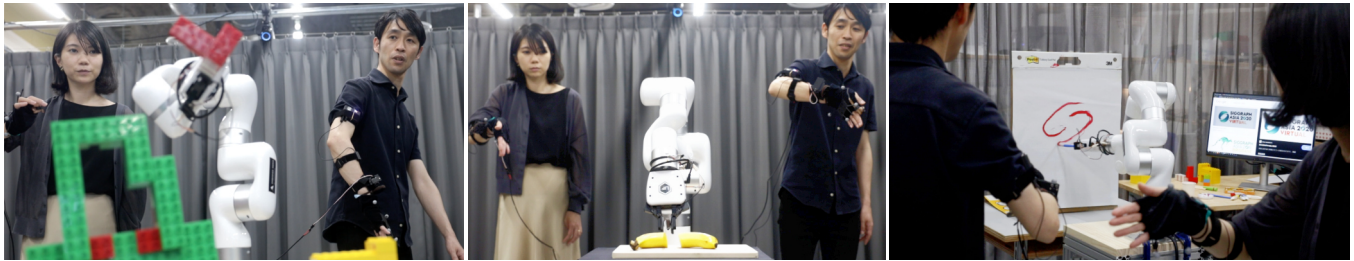


Figure 1: Collaborative avatar platform. Left: division of roles (extended posture); center: fusion (sharing experiences); right: new expressions using collaborative avatar

ABSTRACT

In this study, we developed a system in which two users are integrated into one actual robot arm to collaborate with each other. The method of dividing the roles allows flexible movements that is not dependent on the range of movement and rotational freedom of the human arm, and enables movements that are impossible for a single person. The mixing of movements in adjustable ratio allows an expert to support a beginner to make more stable movements, and induces collaboration between users with different perspectives. Additionally, we implement tactile feedback to enable interaction between users and between users and the robot. We investigate the effects of these on usability and cognitive behavior. This system is expected to become a new method of collaboration in the cyber-physical society.

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1 INTRODUCTION

Our bodies are independent, and there is a one-to-one correspondence between our physical bodies and our perception. However, it is possible to integrate our movement and sensation through robotics and interaction technologies. Integrating the bodies allows us to expand our movements and skills without physical limitations, and to realize diverse co-creation with others.

In the VR space, multiple people can integrate into a single body, and several studies have been conducted on the sense of agency, the sense of body ownership, and the motor performance of the integrated body [Fribourg et al. 2020; Hagiwara et al. 2020]. Fribourg et al. (2020) developed a system in which two users share the arm of an avatar in VR space and manipulate it, and investigated the sense of embodiment for the shared body. The results indicated that the sense of control increased when the goal and movement trajectories of the two users were consistent. Hagiwara et al. (2020) found that

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when a VR avatar shared by two users conducted a reaching task with more smoothness and accuracy than that by a single user. This result suggests that the users prioritized the movement of the shared avatar over their own movement.

For smooth cooperation between humans and robots, mutual recognition is important: the robot recognizes the human behavior and the human recognizes the robot state. To implement this, a system that shares the tactile sensation between human and robot and enables cooperative work was developed [Katayama et al. 2020]. The system indicated that the task completion time was significantly shorter when tactile feedback from the robot was provided.

However, these systems have not indicated physical body integration and sensory connection between humans. In this study, we developed a body integration platform using an actual robot arm and investigated its possibilities. We proposed two control methods that derives different effects to the integrated body and conducted user tests to investigate the potential of those methods.

2 SYSTEM DESIGN

We developed a platform that allows two users to mix and work on one robot arm simultaneously. This system enables movement independent of the physical limitations of the human arm and allows stable collaborative work. The users perform the task in a pair. The movements of two users were measured by a motion capture system (OptiTrack Prime 13W) and processed in a computer. The processed motion data was received by the same computer, and a 7-DoF robot arm (xArm 7) and a gripper (xArm Gripper) was controlled. The velocity feedback and grasping force feedback from the robot arm were provided to the arms of the two users with vibrotactile displays and squeezing displays, respectively. This feedback is promising to recognize the movement of the partner and the environment.

The first control mode is the division of roles between the user who controls the position and the user who controls the posture (Figure 2 top). In the single user situation, physical limitations such as limited degrees of freedom of the joints sometimes make it difficult to work. We found that dividing the position and posture allows us to work with arm twists and angles that are impossible for a single user, thus improving performance (Figure 1 left). Additionally, we found that dividing the roles in each movement allows the users to concentrate on their respective control methods and to divide the focus of their awareness.

The second control mode is the mixing of the movements of the two users into one robot arm (Figure 2 bottom). The position and the angle of the two users' arms are adjusted arbitrarily. In this mode, we found that the movement of the knife was more stable than when a single user was cutting the food (Figure 1 center). This is considered due to unconsciously adjusting their movements according to the movements of the partner by perceiving that they are mixed with the partner's movements. The perception of sharing an avatar with a partner is considered to generate a separate body perception from the normal body perception, and it is expected to adapt the body manipulation method for collaborative work. In addition, receiving tactile feedback from the robot arm which mixed the movement and sensing the existence of the partner through the tactile sense might be one of the factors for optimized movement

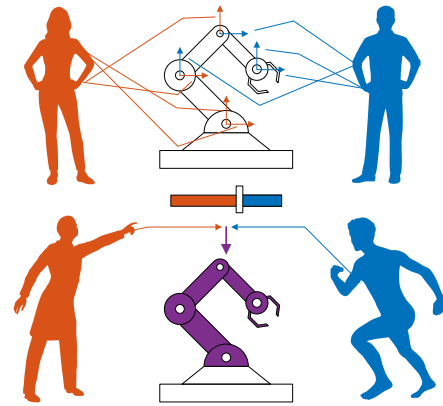


Figure 2: Control mode of an arm avatar. Top: division of roles; bottom: mixed movements

adjustment. This mode allows a beginner to work with the support of an expert. Furthermore, it is possible to receive support from an expert remotely.

In future works, we will quantitatively evaluate the proposed modes and develop seamless switching of the two modes to enhance the integration.

3 APPLICATIONS

The division of roles mode is expected to be used in limited-motion environments, such as repair work on a space station or operation of exploratory robots in the deep sea. Furthermore, it is expected that each person will be able to concentrate on their own role. It also allows experts with different skills in remote places to work through one robot avatar. This allows each user to mix and match their strengths and assist each other's weaknesses, and also to work as a multiple skilled professional. In the mixed movement mode, it is possible to work while perceiving the expert's movement. This mode enables beginners to learn the skills of experts as their own physical sensations, and it is expected to help them learn the skills more efficiently. It is also expected to enhance creativity, such as drawing complex figures that are difficult to draw with a normal body, by two users performing different motions (Figure 1 right).

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